FALL 2024

CSE-4355 Electromechanical Systems and Sensors

THE UNIVERSITY OF TEXAS AT ARLINGTON



LAB 7 REPORT

Submitted toward the partial completion of the requirements for CSE 4355

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Introduction

In this lab, we will be implementing a strain gauge. Strain gauges are used to quantify an object's strain, movement, or deformation. The electrical resistance of the strain gauge changes in proportion to the strain as an object deforms under stress. The amount of strain or applied force can then be determined by measuring this change.

Methods

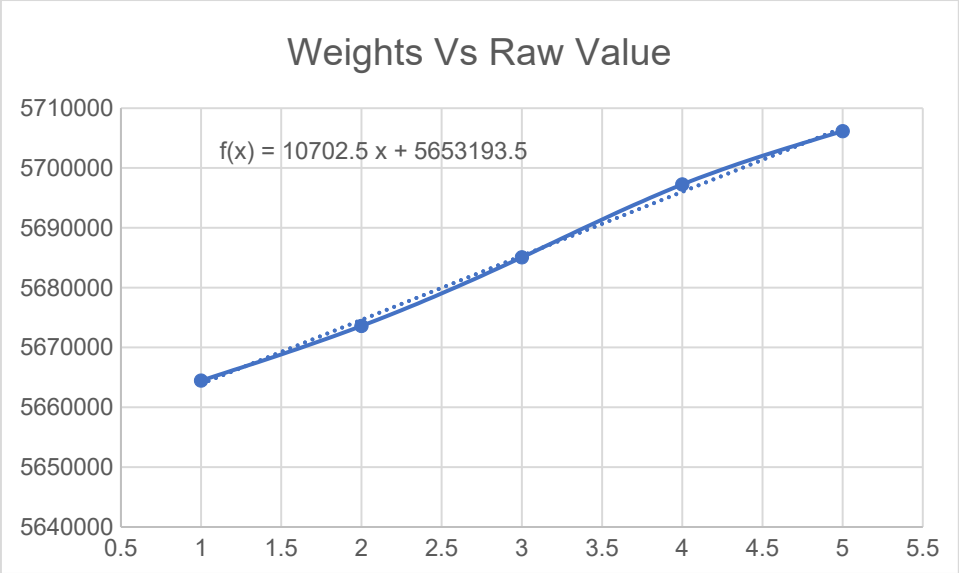
Using the circuit shown in class, we connected the HX711 device's DATA and PD_CLK lines to the controller. We connected 350-ohm fixed value resistors from E+ to A+, E+ to A-, and E- to A+.

We connected E- to the BLACK banana jack and A- to the YELLOW banana jack on the strain gauge block.

1. We soldered 30 AWG bare wires to the strain gauge while it was on a glass plate.
2. We glued the strain gauge to the aluminum beam.
3. We clipped the wires to the BLACK and YELLOW banana jack toothless alligator clips.
4. We configured the clock line as a GPO and the data line as a GPI.
5. We wrote code to read out the value of the voltage across A+ and A- in the Whetstone bridge repeatedly using these steps:
    - Waited for the DATA input to go high, indicating the data was ready to be read.
    - Output 24 clocks, reading in the data bits from MSB to LSB order as follows:
        - Pulled PD_CLK high for less than 50 µs.
        - Read the DATA line and stored the bit in a variable.
        - Pulled PD_CLK low for no less than 200 ns.
        - Repeated until all 24 clocks were sent.
    - Output a 25th clock to indicate that we wanted to sample the A channel with 128 gain.
    - Displayed the value to the UART window.
6. Using the weight from the stepper lab and other items in the lab, we empirically derived the equation for converting the 24-bit A/D result to a force in N and mass in g. We added a display of this force and mass to the loop in step 5.

Initial Experiments

Empirically derived average weight to raw value.



Finger Pressing on Gauge

```
AVERAGE: 6319677
GRAMS: 1896
FORCE (NEWTONS): 193338
GRAMS: 1919
FORCE (NEWTONS): 195683
GRAMS: 1932
FORCE (NEWTONS): 197009
GRAMS: 1890
FORCE (NEWTONS): 192726
GRAMS: 1804
FORCE (NEWTONS): 183956
GRAMS: 1744
FORCE (NEWTONS): 177838
GRAMS: 1715
FORCE (NEWTONS): 174881
GRAMS: 1670
FORCE (NEWTONS): 170292
GRAMS: 1687
FORCE (NEWTONS): 172026
GRAMS: 1700
FORCE (NEWTONS): 173351

AVERAGE: 6398700
GRAMS: 1700
FORCE (NEWTONS): 173351
GRAMS: 1700
FORCE (NEWTONS): 173351
GRAMS: 1700
FORCE (NEWTONS): 173351
```

Below are different test ran with the sensor.

No Weights:

```
AVERAGE: 6331996
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
```

3 W

```
AVERAGE: 6299908
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 549
FORCE (NEWTONS): 55982
GRAMS: 685
FORCE (NEWTONS): 69850
GRAMS: 679
FORCE (NEWTONS): 69238
GRAMS: 672
FORCE (NEWTONS): 68524
GRAMS: 674
FORCE (NEWTONS): 68728
GRAMS: 674
FORCE (NEWTONS): 68728
GRAMS: 673
FORCE (NEWTONS): 68626
GRAMS: 675
FORCE (NEWTONS): 68830
GRAMS: 672
FORCE (NEWTONS): 68524

AVERAGE: 6329375
GRAMS: 672
FORCE (NEWTONS): 68524
GRAMS: 672
FORCE (NEWTONS): 68524
GRAMS: 672
FORCE (NEWTONS): 68524
GRAMS: 672
FORCE (NEWTONS): 68524
GRAMS: 672
```

1 Weight on Sensor

```
AVERAGE: 6315639
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 404
FORCE (NEWTONS): 41196
GRAMS: 324
FORCE (NEWTONS): 33038
GRAMS: 293
FORCE (NEWTONS): 29877
GRAMS: 276
FORCE (NEWTONS): 28144
GRAMS: 252
FORCE (NEWTONS): 25696
GRAMS: 248
FORCE (NEWTONS): 25288
GRAMS: 243
FORCE (NEWTONS): 24779
GRAMS: 238
FORCE (NEWTONS): 24269
GRAMS: 235
FORCE (NEWTONS): 23963

AVERAGE: 6327848
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
```

Removing weig

```
7848
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 235
FORCE (NEWTONS): 23963
GRAMS: 0
FORCE (NEWTONS): 0

AVERAGE: 6322758
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
GRAMS: 0
FORCE (NEWTONS): 0
```

Conclusions

There were many difficulties working with the sensor because of the drift of the sensor. Because of this drift

we had to had implement a software solution which ended up being an attempt at a sliding average window and taking the difference when there is a large spike. An additional strain gauges could be potentially added to the opposite or bottom side of the rod to eliminate this. Some limitations with our implementation included having to remove and put all weight at one time and not being able to weigh something lighter than a weight. The latter was due to everything being calibrated from the weights provided in the lab. More time would allow for these issues to be resolved.

In conclusion, we learned how to attach and implement strain gauges in this lab. The surface of the object being measured is usually where strain gauges are mounted. The strain gauge's electrical resistance varies in proportion to changes in the object's size. An electrical signal is subsequently produced from this resistance change.

Circuit Built:

